United States Patent Application Publication 20250276623
Kind Code A1
Publication Date September 04, 2025
Inventor(s) SCHUKALSKI; Jürgen

CRASH DEVICE FOR A VEHICLE SEAT

Abstract

It is provided a crash device, in particular for a vehicle seat, comprising: a first and a second bearing point for connecting one component each, and a carrier on which the first bearing point is formed and which includes a guide for guiding a movement of the second bearing point relative to the first bearing point, wherein in a starting position the second bearing point is arrested by an arresting element which is locked by means of a locking device and after unlocking of the locking device can be plastically deformed by a movement of the second bearing point along the guide.

Inventors: **SCHUKALSKI; Jürgen (Küps, DE)**

Applicant: **Brose Fahrzeugteile SE & Co. Kommanditgesellschaft, Coburg** (Coburg, DE)

Family ID: **96738154**

Appl. No.: **19/054607**

Filed: **February 14, 2025**

Foreign Application Priority Data

DE	10 2024 105 888.9	Feb. 29, 2024

Publication Classification

Int. Cl.: **B60N2/427** (20060101)

U.S. Cl.:

CPC **B60N2/42781** (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to German Patent Application No. 10 2024 105 888.9 filed on Feb. 29, 2024. The entire contents of the above-listed application are hereby incorporated by reference for all purposes.

BACKGROUND

[0002] The proposed solution relates to a crash device and a vehicle seat comprising such a crash device.

[0003] In a vehicle crash large forces can act on a vehicle seat and the passenger sitting thereon. DE 10 2021 202 560 A1 describes a lever with a deformation section, which in a vehicle crash can reduce loads due to a deformation of the deformation section and thus provides for slowing down the passenger more gently.

[0004] Furthermore, vehicle seats can be adjustable, e.g. in order to provide each of various seat users with a comfortable seating position and to be adapted to various spatial conditions. For example, a vehicle seat can be mounted in a vehicle so as to be longitudinally adjustable by means of a longitudinal adjustment device, in order to provide for various seating positions along the longitudinal vehicle axis, a vehicle seat can be adjustable by means of a height adjustment device, in order to adjust a seat height, and in a vehicle seat the inclination of a backrest can be adjustable relative to a seat part, in order to name only a few examples. The lever described in DE 10 2021 202 560 A1 is part of a seat height adjustment of the vehicle seat and hence fulfills a dual purpose.

[0005] To provide a particularly comfortable seating position for passengers, and in autonomously driving vehicles also for drivers, during the ride, vehicle seats can be designed such that they can be transferred into a relax position in which a seat part and a backrest are inclined far to the rear. In such a position, which in particular can represent a reclined position, the passenger can lie and e.g. sleep comfortably. In such a position of the vehicle seat, however, high forces would be introduced into the spine in the case of a front crash due to the restraint of the pelvis of the seat user. Depending on the concrete configuration of the vehicle seat, this can result in an increased risk of injury of the seat user. In such a seat position, the reduction of crash energy and/or the transfer into a position to be cushioned more easily therefore can be particularly advantageous.

SUMMARY

[0006] It is the object to indicate a crash device improved further.

[0007] This object is achieved by a crash device having features as described herein.

[0008] Accordingly, there is indicated a crash device, in particular for a vehicle seat. The crash device comprises a first bearing point and a second bearing point, each for connection of a respective component (external with respect to the crash device), for example of a respective component of the vehicle seat. The crash device furthermore comprises a carrier, on which the first bearing point is formed and which includes a guide for guiding a movement of the second bearing point relative to the first bearing point. It is provided that in a starting position the second bearing point is arrested by an arresting element which is locked by means of a locking device and after unlocking of the locking device can be deformed (in particular plastically) by a movement of the second bearing point along the guide.

[0009] By means of the locking device, a movement of the second bearing point along the guide can selectively be prevented or enabled. In this way, the locking device can be unlocked e.g. in dependence on the seating position, the acting acceleration or the detection of a vehicle crash, so that it is possible to choose the behavior of the crash device optimal for the respective situation. In a vehicle seat comprising one or more of such crash devices, the vehicle seat thereby can be moved into another position, e.g. into a position in which the remaining forces can be absorbed more easily. Concretely the vehicle seat can be transferred e.g. from the relax position (or another, in particular rearwardly inclined starting position) into a more upright position by comparison, so that the acting forces can be introduced over a larger surface area via a seat belt and a risk of “submarining”, i.e. slipping through under the seat belt, can be reduced distinctly. The load acting on the spine can be reduced thereby. One of the bearing points or both bearing points can be a pivot bearing or a part of a pivot bearing, wherein another kind of support also each is conceivable, such as an immovable, fixed support on one of the bearing points.

[0010] Since the arresting element is deformable in order to enable the movement (and e.g. a deformation of a deformation element, see below), a clearance-free support of the second bearing portion also is made possible. For example, no clearance is necessary for a rotatable or shiftable support of the arresting element or the like. As a result, an improved crash device thus is achieved and consequently an improved vehicle seat is made possible. In addition, an easy assembly is made possible, for example only a joining method is carried out, e.g. welding.

[0011] The crash device furthermore can comprise a deformation element which can be deformed by a movement of the second bearing point guided on the guide out of the starting position relative to the first bearing point. By means of the locking device, a deformation of the deformation element selectively can be prevented or enabled. For example in the case of a frontal impact of the vehicle, crash energy can be reduced by means of the deformation element in that material of the deformation elements is elastically and/or plastically deformed. The crash device hence can be a crash absorber.

[0012] It can be provided that the locking device and/or the arresting element each are mounted on the carrier in such a way that they cannot be moved without being destroyed. A force acting on the arresting element can be introduced for example via a first load path from the arresting element (in particular directly, e.g. via a contact between the arresting element and the carrier) into the carrier. The force can also be introduced via a second load path from the arresting element via the locking device in the locked state into the carrier. The arresting element can be weakened by opening the second load path by means of unlocking of the locking device. The locking device and the arresting element hence can be configured as non-movable parts, and the arresting element can be weakened in a targeted way by removing the second load path, in particular can be deformed in a targeted way (e.g. when the acting force exceeds a threshold value), instead of opening e.g. via a movable support. After unlocking of the locking device and the deformation of the arresting element, the same hence e.g. can no longer be transferred back into the locked starting position, as they are destroyed due to the plastic deformation.

[0013] The arresting element can be attached to the carrier, e.g. be welded to the carrier. In this way, a clearance between the arresting element and the carrier can be avoided.

[0014] It can be provided that the arresting element includes an area (e.g. arm) which encloses the second bearing point in the starting position. Furthermore, it can be provided that the arresting element can be deformed elastically and/or plastically due to the movement of the second bearing point along the guide. Such an area provides for a safe arrestment in the starting position and a controlled release of the bearing point in the case of a crash.

[0015] The locking device can comprise a locking element fixed to the carrier. It can be provided that the locking element in a locking position is in operative connection, in particular in engagement, especially in contact with the arresting element and/or can be transferred into an unlocking position (in particular by plastic deformation), in which the locking element is out of operative connection and/or engagement and/or contact with the arresting element. The arresting element thereby can be locked in a simple way in the position securing the second bearing point in the starting position. Unlocking can be effected with little expenditure of force and the arresting element yet can be safely protected against unwanted opening. By eliminating the operative connection or the engagement or the contact of the locking element, the second load path can be eliminated. The first load path is overloaded e.g. from a particular load such that the arresting element is deformed in a targeted way and the bearing point can move.

[0016] The locking element for example includes a receptacle. In the locking position, the receptacle can e.g. be positively in engagement with the area (e.g. arm) of the arresting element, in

particular with an end of the area (e.g. arm). With a simple and robust construction, this provides for safe locking.
[0017] For example, the locking element can be transferred from the locking position into the unlocking position by a deformation of the locking element. Thus, a movable support of the locking element can (also) be omitted. A clearance in the support of the locking element can thereby be avoided. It can even be provided that in the locking position the locking element is pressed onto the area (e.g. arm) of the arresting element.
[0018] In some embodiments, the locking device comprises a trigger element. By means of the trigger element, the locking element can be transferable from the locking position into the unlocking position. In this way, a needs-based release of the deformation of the deformation element can be controlled.
[0019] The trigger element is configured e.g. in the form of a pyrotechnical actuator. The trigger element can comprise an igniter and a propellant charge ignitable by the igniter. This provides for a particularly fast release of the deformation element.
[0020] The guide defines a guideway, e.g. a linear or curved guideway, along which the movement of the second bearing point out of the starting position is guided relative to the first bearing point. It can be provided that the trigger element is adapted to exert a force on the locking element at an angle to the guideway. Such an arrangement allows safe locking. The angle can be a right angle. It can thus be prevented that even larger forces acting on the second bearing point do not lead to a deformation of the deformation element, as long as the locking device is not unlocked.
[0021] In some embodiments, the carrier forms a housing. In the housing, the arresting element can be arranged. The arresting element thus can be protected against external influences.
[0022] The carrier includes e.g. two housing sheets which can be welded to each other. Thus, a construction can be achieved, which can be manufactured particularly easily and is robust at the same time.
[0023] For example, the guide is configured in the form of an oblong hole and/or is formed in at least one of the housing sheets. This allows a particularly safe guidance.
[0024] It can be provided that the second bearing point comprises a bearing bush and/or a pin. In the starting position, the bush (or the pin) can be pressed into the arresting element. Thereby, a completely clearance-free support can be achieved.
[0025] The first bearing point and/or the second bearing point (each) can directly or indirectly be formed for a rotatable support. The crash device can be used e.g. as a swingarm in a height adjustment device or the like and thus fulfill a dual function.
[0026] Unlocking of the locking device e.g. is irreversible. This provides for a particularly simple and safe construction.
[0027] According to one aspect, there is indicated a vehicle seat (e.g. with a seat part and a backrest), comprising one or more crash devices according to any of the embodiments described herein. As regards the advantages, reference is made to the above indications.
[0028] It can be provided that the seat part (as one of the components) is supported on a base (as the other one of the components) via the crash device(s). Via the crash device(s), e.g. a weight of the seat part is supported on the base. The crash device (or the crash devices) thus can fulfill a dual function.
[0029] The (respective) crash device can be pivotally mounted on the base with one of the two bearing points. The seat part can be pivotally mounted on the other one of the bearing points. The crash device or the crash devices thus can be used e.g. as swingarms of an adjustment device.
[0030] It can be provided that the seat part is height-adjustably mounted relative to the base via the crash device(s). In normal use, the respective crash device hence can be used as swingarms of a height adjustment device and reduce crash energy in the case of a crash.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] The idea underlying the solution will be explained in detail below with reference to the exemplary embodiments illustrated in the Figures.

[0032] FIG. **1** shows an adjustable vehicle seat with a seat part and a backrest.

[0033] FIG. **2** shows a crash device of the vehicle seat of FIG. **1** with a deformation element that can be deformed as a result of an action of force.

[0034] FIG. **3**A shows a sectional view of the crash device of FIG. **2** in a starting position according to the sectional plane depicted in FIG. **3**B.

[0035] FIG. **3**B shows a side view of the crash device of FIG. **2** in the starting position.

[0036] FIG. **4**A shows a sectional view of the crash device of FIG. **2** in a starting position according to the sectional plane depicted in FIG. **4**B.

[0037] FIG. **4**B shows a side view of the crash device of FIG. **2** in the starting position.

[0038] FIG. **5**A shows a sectional view of the crash device of FIG. **2** according to the sectional plane depicted in FIG. **5**B.

[0039] FIG. **5**B shows a side view of the crash device of FIG. **2**, wherein a locking device is unlocked and releases an arresting element of the crash device.

[0040] FIG. **6**A shows a sectional view of the crash device of FIG. **2** according to the sectional plane depicted in FIG. **6**B.

[0041] FIG. **6**B shows a side view of the crash device of FIG. **2**, wherein the arresting element of the crash device has been deformed due to a movement of a second bearing point of the crash device along a guide.

[0042] FIG. **7** shows a further crash device for the vehicle seat of FIG. **1**.

[0043] FIGS. **8** and **9** show trigger elements for the crash device of FIG. **2**.

[0044] FIG. **10** shows the deformation element of the crash device of FIG. **2**.

DETAILED DESCRIPTION

[0045] FIG. **1** shows a vehicle seat **2** comprising a seat part **20** and a backrest **21**. The backrest **21** is arranged on a rear area of the seat part **20**, in the present case pivotally mounted on the seat part **20** about a pivot axis by means of an arrangement of fittings **23**.

[0046] The vehicle seat **2** furthermore comprises a height adjustment device **24** for adjusting the seat height of the seat part **20** (along with the backrest **21**) relative to a base **22**. The height adjustment device **24** (on each side, i.e. on the left and right) in a front area includes a front swingarm **240** and in a rear area a rear swingarm **241**, via which the seat part **20** and here also the backrest **21** are supported on the base **22**. The swingarms **240**, **241** each are pivotally mounted on the base **22**, and the seat part **20** is pivotally mounted on the swingarms **240**, **241**.

[0047] By adjusting the swingarms **240**, **241** relative to the base **22**, the seat height can be adjusted relative to the base **22** along a vertical vehicle axis Z.

[0048] In the present case, a seat rail **250** of a longitudinal adjustment device **25** of the vehicle seat **2** serves as a base **22**. When the vehicle seat **2** e.g. is not equipped with the longitudinal adjustment device **25**, e.g. a vehicle floor **3** or a component fixed thereto can serve as a base.

[0049] By means of the longitudinal adjustment device **25** the seat part **20** (jointly with the backrest **21**) can be adjusted relative to the vehicle floor **3** along a longitudinal vehicle axis X. The longitudinal vehicle axis X extends perpendicularly to the vertical vehicle axis Z. The longitudinal vehicle axis X and the vertical vehicle axis Z each extend perpendicularly to a transverse vehicle axis Y. The pivot axes of the swingarms **240**, **241** extend parallel to the transverse vehicle axis Y. The pivot axis of the backrest **21** extends parallel to the transverse vehicle axis Y relative to the seat part **20**.

[0050] In the present example, the longitudinal adjustment device **25** comprises two floor rails **251** spaced along the transverse vehicle axis Y, of which in the side view of FIG. **1** the left floor rail **251** from the point of view of a seat user sitting on the vehicle seat **2** can be recognized. The corresponding right floor rail **251** is formed analogously to the left floor rail **251** (e.g. identically or mirror-symmetrically). This also applies for the swingarms **240**, **241** of the height adjustment device **24** mentioned already.

[0051] The floor rails **251** can be mounted on the vehicle floor **3** of a vehicle and, in the mounted stated as shown in FIG. **1**, are attached to the vehicle floor **3** of the vehicle comprising the vehicle seat **2**. A respective seat rail **250** of the longitudinal adjustment device **25** is in engagement with each of the floor rails **251** so as to be longitudinally shiftable. The longitudinal adjustment device **25** connects the height adjustment device **24** to the vehicle floor **3**.

[0052] FIG. **1** shows the vehicle seat **2** in an upright seating position, in which the seat user can drive the vehicle with the vehicle seat **2** e.g. as a driver. The vehicle seat **2** can also be adjusted into a relax position, in which the seat user can e.g. rest or sleep in the vehicle seat **2**. In the relax position both the seat part **20** and the backrest **21** are inclined further to the rear as compared with the upright seating position (from the point of view of the seat user sitting on the vehicle seat **2** and looking straight ahead), namely in the present case to such an extent that a torso angle and/or an angle of a longitudinal axis of the backrest **21** with an alignment of the base **22** according to a mounted state are more than 25° (in particular more than 30°, more than 40° or even more than 50°) relative to the vertical vehicle axis Z (and in the vehicle placed on a horizontal ground relative to the vertical Z). The torso angle by way of example corresponds to the angle of a straight torso line of the torso from the hip joint to the shoulders of a seat user sitting in the vehicle seat **2** relative to the vertical vehicle axis Z (and/or relative to the vertical Z). The longitudinal axis of the backrest for example extends from the pivot axis of the backrest **21** on the seat part **20** up to the upper end edge of the backrest **21**.

[0053] The measurement of the torso angle for example is effected with the standardized H-point measuring machine SAE J826 H-POINT MANIKIN according to E/ECE/324 and/or E/ECE/TRANS/505 (e.g. according to Regulation No. 14, REV. 1/ADD. 13/REF, 1, ANNEX 4).

[0054] The seat part **20** and the backrest **21** can be moved by means of the height adjustment device **24** from the upright seating position into the relax position inclined further to the rear, and vice versa.

[0055] In the present example, a belt exit point of a seat belt **26** for the seat user furthermore is arranged at an upper end (facing away from the seat part **20**) of the backrest **21**. A winding mechanism for the seat belt **26** is mounted e.g. on the backrest **21**. By pulling on the seat belt **26** a force hence acts on the upper end of the backrest **21**. Furthermore, a belt buckle is attached to the seat part **20** (alternatively e.g. to the seat rail **250**).

[0056] In the case of a front crash, the weight of the seat user pulls on the backrest **21** via the seat belt **26** and exerts a force on the backrest **21**. Furthermore, a force acts on the seat part **20** in the region of the belt buckle. The forces each are directed forwards and upwards. As in the relax position the torso of the seat user is inclined relatively far to the rear, e.g. at an angle of more than 35° relative to the vertical vehicle axis Z or even at an angle of more than 45° relative to the vertical vehicle axis Z, high forces would be introduced into the spine of the seat user in the case of a frontal impact.

[0057] The front swingarms **240** (alternatively or additionally the rear swingarms **241**) therefore are configured in the form of crash devices **1**A, as illustrated in FIG. **2**. The front suspension of the seat part **20** thereby can be compressed, whereby in the case of a crash of the vehicle a movement of the seat part **20** and the backrest **21** comprising a rotation is guided from the relax position (or another position) into a comparatively more upright position.

[0058] As will be explained in more detail below, energy introduced into the vehicle seat **2** as a result of the crash is absorbed during this movement from the relax position into the more upright position. It should be noted already at this point that the described arrangement of the crash devices

1A merely is by way of example, and other or additional arrangements of such crash devices **1**A on the vehicle seat **2** (e.g. on the longitudinal adjustment device **25** and/or on the fittings **23**) or at some other point also are conceivable.

[0059] FIG. **2** shows one of the crash devices **1**A of the vehicle seat **2**. The crash device **1**A can also be referred to as a deformable connecting member or as a holding part, according to the present example also as a swingarm. In the present example, the crash device **1**A forms a crash absorber (for the absorption of crash energy).

[0060] The crash device **1**A comprises a first bearing point **14** and a second bearing point **15**, each for connection of a component, in the present case for the pivotable connection of the seat part **20** at one of the bearing points **14**, **15** and of the base **22** at the other one of the bearing points **14**, **15**. The bearing points **14**, **15** therefor each comprise e.g. a bolt and/or, as shown in FIG. **2**, an opening **141**, **152**. In the present case, merely by way of example, the opening **141** of the first bearing point **14** formed as a through opening is provided with a sleeve **140** into which e.g. a bolt or the like can be plugged in order to be rotatably mounted in the sleeve **140**. Likewise by way of example, the second bearing point **15** is equipped with a through opening into which a bolt or the like can be plugged. However, other embodiments also are possible as well. In the present case, the first bearing point **14** and the second bearing point **15** each are configured for rotatably mounting the respectively connected component. With the components (seat part **20** and base **22**) connected according to FIG. **1**, the two bearing points **14**, **15** each form a pivot bearing.

[0061] The crash device **1**A furthermore comprises a carrier **10** on which the first bearing point **14** is formed and which includes a guide **100** for guiding a movement of the second bearing point **15** relative to the first bearing point **14**. The second bearing point **15** is mounted on the guide **100** so as to be shiftable along the guide **100**. FIG. **2** shows the second bearing point **15** in a starting position. In the starting position, the second bearing point **15** is arranged at the end of the guide **100** facing away from the first bearing point **14**.

[0062] The carrier **10** comprises two housing sheets **102**, **103** welded to each other. In the present case, the housing sheets **102**, **103** by way of example are stamped and bent parts. The guide **100** is configured in the form of an oblong hole in at least one of the housing sheets **102**, **103**, in the present case in the upper housing sheet **103**.

[0063] The crash device **1**A furthermore comprises a deformation element **11**. The deformation element **11** can be deformed, in the present case plastically, by a movement of the second bearing point **15** out of the starting position relative to the first bearing point **14**, which is guided on the guide **100**. In the illustrated example, the second bearing point **15** can be moved towards the first bearing point **14**, wherein the deformation element **11** is deformed, as will yet be explained further below.

[0064] It is provided that in the starting position the second bearing point **15** is arrested by an arresting element **12** which is locked by means of a locking device **13** and which after unlocking of the locking device **13** can be deformed by a movement of the second bearing point **15** along the guide **100** (simultaneously with the deformation element **11** or before the deformation of the deformation element **11**).

[0065] In the illustrated example, the guide **100** is oblong. The guide **100** defines a guideway **101**, which is linear according to FIG. **2**, but might also be formed differently, e.g. arc-shaped, in particular according to a circular segment. The deformation element **11** is attached to the guide **100**. The deformation element **11** extends along the guide **100**. The deformation element **11** blocks the guide **100**. In the guide **100**, the second bearing point **15** is arranged.

[0066] With respect to FIGS. **3**A to **6**B the function of the arresting element **12** and the locking device **13** will now be explained in detail.

[0067] FIGS. **3**A to **4**B show the starting position. The arresting element **12** encloses the second bearing point **15**, namely in the present example almost once around its axis. The arresting element **12** includes a base **122** and an area, here by way of example in the form of an arm **120**. The base

122 is arranged at an end of the crash device **1**A facing away from the first bearing point **14**. The base **122** of the arresting element **12** is arranged on the side of the second bearing point **15** which faces away from the first bearing point **14**.
[0068] The arresting element **12** is arranged between the two housing sheets **102**, **103**. The housing sheets **102**, **103** jointly form a housing for the arresting element **12**, the deformation element **11** and some parts of the locking device **13**. The arresting element **12** is firmly held at the carrier **10**. In the present case, the arresting element **12** is fixed to the carrier **10**, namely concretely welded thereto.
[0069] The arm **120** of the arresting element **12** extends around the second bearing point **15** and through an opening **104** of the carrier **10**. In the present case, the opening **104** is formed in the upper housing sheet **103**, namely in the present case in a side wall. An open end **121** of the arm **120** of the arresting element **12** is arranged outside the carrier **10**. The arm **120** of the arresting element **12** blocks a movement of the second bearing point **15** along the guide **100**. The arm **120** of the arresting element **12** arrests the second bearing point **15** in the starting position. As long as the arm **120** is in the position shown in particular with reference to FIG. **4**A, a high (higher) force can be transmitted (than with an unlocked arresting element **12**), without the second bearing point **15** being shifted along the guide **100**.
[0070] To hold the arm **120** in this position, the arm **120** is held by the locking device **13**. For this purpose, the locking device **13** comprises a locking element **130**. The locking element **130** is fixed to the carrier **10**. For this purpose, the locking element **130** includes a fastening area **135** which is attached to the carrier **10**. In the present example, the fastening area **135** is welded to the carrier **10**. The locking element **130** furthermore includes a receptacle **134**. The end **121** of the arm **120** of the arresting element **12** is plugged into the receptacle **134**. The direction along which the end **121** of the arm **120** of the arresting element **12** is plugged into the receptacle **134** is aligned perpendicularly to the guideway **101**.
[0071] In the position shown in FIG. **4**A, the second bearing point **15** is pressed (between base **122** and arm **120**) into the arresting element **12** (e.g. without any clearance) and positively and non-positively held therein. The second bearing point **15** includes a bearing bush **150** (alternatively or additionally a pin), which here is plugged, preferably pressed into the arresting element **12**, and here by way of example is of circularly cylindrical shape. Furthermore, the end **121** of the arm **120** of the arresting element **12** is pressed into the receptacle **134** in such a way that it rests against a side surface of the receptacle **134** under tension. The bearing bush **150** by way of example here includes an internal thread into which a bolt can be screwed. This bolt can form a pivot bearing. Alternatively or additionally, the bearing bush **150** can rotatably support the bolt.
[0072] In its locking position, the locking element **130** is in engagement with the arresting element **12** and can be transferred into an unlocking position in which it is out of engagement with the arresting element **12**. The unlocking position is illustrated in FIGS. **5**A and **5**B. Due to a deformation, the locking element **130** here has been transferred from the locking position into the unlocking position. In the process, a web **136** on which the receptacle **134** is formed is pivoted away from the carrier **10**, namely in the present case bent up. Between the web **136** and the fastening area **135** a weakened portion of the locking element **130** is provided, at which the locking element **130** in the present case is bent up in the unlocking position.
[0073] To transfer the locking element **130** from the locking position into the unlocking position, the locking device **13** comprises a trigger element **131**A. The trigger element **131**A is aligned perpendicularly to the guideway **101**. The trigger element **131**A is arranged in the housing formed by the carrier **10**. In the present case, the trigger element **131**A is arranged adjacent to the second bearing point **15**. The trigger element **131**A extends adjacent to the arm **120** of the arresting element **12**. In the present case, the trigger element **131**A is aligned parallel to the arm **120**. The trigger element **131**A is plugged into a side wall of the carrier **10** and held on the same. A tip of the trigger element **131**A is aligned with the locking element **130**, namely in the present case concretely with a protrusion **137** which is extended into the opening **104** of the carrier **10** towards

the trigger element **131**A.

[0074] In the present example, the trigger element **131**A is formed as shown in FIG. **8**.

[0075] As shown in FIG. **8**, the trigger element **131**A is configured in the form of a pyrotechnical actuator with an igniter **133** and a propellant charge **132** to be ignited by the igniter **133**. At the tip of the trigger element **131**A, as shown, a predetermined breaking point can be formed. On ignition of the propellant charge **132** the trigger element **131**A breaks up, concretely at the predetermined breaking point, and with combustion gases and/or a housing part and/or an optionally provided stamp or the like exerts a force F (see FIG. **5**A) on the locking element **130**, concretely on its protrusion **137**. The locking element **130** thereby bends plastically (alternatively or additionally elastically) away from the carrier **10** and thus gets out of engagement with the arm **120** of the arresting element **12**. The force F is aligned perpendicularly to the guideway **101**.

[0076] At this point it should be noted that instead of the pyrotechnical actuator a differently configured trigger element can also be provided, e.g. the trigger element **131**B as shown in FIG. **9**. Accordingly, a spring accumulator with a pretensioned spring **139** is arranged in the interior of the trigger element **131**B. A latch **138** or the like holds the spring **139** in the pretensioned position. An actuation of the latch **138** (here by pulling out) releases the spring **139** so that the same expands and exerts a force F on the locking element **130**, which thereby releases the arm **120** of the arresting element **12** as described.

[0077] Unlocking of the locking device **13** consequently is irreversible, which provides for a simple and robust construction.

[0078] In the position shown in FIG. **5**A, the arm **120** of the arresting element **12** thus still is arranged in the starting position, but can now be deformed (plastically), namely be bent up, by a force F directed towards the first bearing point **14** onto the second bearing point **15**, see in particular FIG. **6**A.

[0079] Here (or depending on the arrangement of the deformation element **11** thereafter) the deformation element **11** also is deformed and the second bearing point **15** is shifted along the guide **100**. A slide **151** of the second bearing point **15** is in planar abutment with the edges of the guide **100**. The arresting element **12** cannot be opened without being destroyed.

[0080] The locking device **13** and the arresting element **12** each are mounted on the carrier **10** in such a way that the locking element **130** and the area (here the arm **120**) cannot be moved without being destroyed. A force on the arresting element **12** can directly be introduced into the carrier **10** via a first load path from the arresting element **12** and can be introduced into the carrier **10** via a second load path from the arresting element **12** via the locking device **13** disposed in the locked state. The arresting element **12** can be weakened by removing the second load path by means of unlocking of the locking device **13**.

[0081] The trigger element **131**A, **131**B can be triggered by a control system. The control system can detect an imminent or effected crash and in response thereto activate the trigger element **131**A, **131**B. The control system optionally detects a position of the vehicle seat **2** and activates the trigger element **131**A, **131**B only in a specified position of the vehicle seat. The trigger element **131**A, **131**B can also be triggerable by an acting acceleration. The actuation of the trigger element **131**A, **131**B can be effected via a seat-integrated crash assessment, an acceleration sensor or an on-board crash assessment, and by detecting an imminent vehicle crash or by detecting a vehicle crash occurred already.

[0082] FIG. **7** shows a crash device **1**B which the vehicle seat **2** can comprise alternatively or in addition. The mode of operation is the same as described above, merely the direction of the movement of the second bearing point **15** in the guide **100** is different, namely away from the first bearing point **14**. The carrier **10** includes the correspondingly configured guide **100**. Thus, the distance between the bearing points **14**, **15** can be increased.

[0083] FIG. **10** shows the deformation element **11** separately in an exemplary embodiment. The deformation element **11** here includes weakened portions **110**. The deformation element **11** is

formed weaker than the housing sheets **102**, **103**. The weakened portions **110** can be configured in the form of apertures or indentations. The deformation element **11** allows a particularly precise and uniform deformation. The deformation element **11** can have a smaller material thickness and/or a material with a lower strength than the carrier **10**. By action of a force on the second bearing point **15** in the guide **100** along the guide **100**, which exceeds a predetermined minimum force, the deformation element **11** fails and is deformed, plastically in the example described here. The deformation element **11** thereby releases the guide **100** and permits a displacement of the second bearing point **15** along the guide **100**. In the starting position of the second bearing point **15**, the guide **100** is concealed by the deformation element **11**. The deformation element **11** is oblong.
[0084] Optionally, the deformation element **11** has an (e.g. linearly, incrementally or exponentially) rising material thickness, strength and/or stability along its length. For example, people of different weights can be safely trapped. Before an end position in the guide **100**, such a progression can be provided alternatively or in addition in order to prevent a hard impact.
[0085] Alternatively or additionally, the deformation element might also be formed by opposing edges of the guide **100**. By exceeding the minimum force, the second bearing point **15** can bend the edges and thus provide for the displacement in the guide **100**.
LIST OF REFERENCE NUMERALS
[0086] **1**A, **1**B crash device [0087] **10** carrier [0088] **100** guide [0089] **101** guideway [0090] **102**, **103** housing sheet [0091] **104** opening [0092] **11** deformation element [0093] **110** weakened portion [0094] **12** arresting element [0095] **120** area (arm) [0096] **121** end [0097] **122** base [0098] **13** locking device [0099] **130** locking element [0100] **131**A, **131**B trigger element [0101] **132** propellant charge [0102] **133** igniter [0103] **134** receptacle [0104] **135** fastening area [0105] **136** web [0106] **137** protrusion [0107] **138** latch [0108] **139** spring [0109] **14** first bearing point [0110] **140** sleeve [0111] **141** opening [0112] **15** second bearing point [0113] **150** bearing bush [0114] **151** slide [0115] **152** opening [0116] **2** vehicle seat [0117] **20** seat part (component) [0118] **21** backrest [0119] **22** base (component) [0120] **23** fitting [0121] **24** height adjustment device [0122] **240**, **241** swingarm [0123] **25** longitudinal adjustment device [0124] **250** seat rail [0125] **251** floor rail [0126] **26** seat belt [0127] **3** vehicle floor [0128] F force [0129] X longitudinal vehicle axis [0130] Y transverse vehicle axis [0131] Z vertical vehicle axis

Claims

1. A crash device comprising: a first and a second bearing point for connecting one component each, and a carrier on which the first bearing point is formed and which includes a guide for guiding a movement of the second bearing point relative to the first bearing point, wherein in a starting position the second bearing point is arrested by an arresting element which is locked by means of a locking device and after unlocking of the locking device can be plastically deformed by a movement of the second bearing point along the guide.
2. The crash device according to claim 1, further comprising a deformation element which can be deformed relative to the first bearing point by a movement of the second bearing point guided on the guide out of a starting position.
3. The crash device according to claim 1, wherein the locking device and the arresting element each are mounted on the carrier in such a way that they cannot be moved without being destroyed, wherein a force acting on the arresting element can directly be introduced into the carrier via a first load path from the arresting element and via the locking device in the locked state can be introduced into the carrier via a second load path from the arresting element, and wherein by removing the second load path the arresting element can be weakened by means of unlocking of the locking device.
4. The crash device according to claim 1, wherein the arresting element is firmly attached to the carrier.

5. The crash device according to claim 1, wherein the arresting element includes an area which encloses the second bearing point in the starting position and can be plastically deformed by the movement of the second bearing point along the guide.
6. The crash device according to claim 1, wherein the locking device comprises a locking element fixed to the carrier, which in a locking position is in operative connection with the arresting element and can be transferred into an unlocking position in which the locking element is out of operative connection with the arresting element.
7. The crash device according to claim 5, wherein the locking device comprises a locking element fixed to the carrier, which in a locking position is in operative connection with the arresting element and can be transferred into an unlocking position in which the locking element is out of operative connection with the arresting element, the locking element includes a receptacle which in the locking position is in operative connection with an end of the area of the arresting element.
8. The crash device according to claim 6, wherein the operative connection of the locking element with the arresting element can be released by a deformation of the locking element.
9. The crash device according to claim 6, wherein the locking device comprises a trigger element by means of which the locking element can be transferred from the locking position into the unlocking position.
10. The crash device according to claim 9, wherein at least one of the trigger element is configured in the form of a pyrotechnical actuator with an igniter and a propellant charge to be ignited by the igniter; and the guide defines a guideway along which the movement of the second bearing point out of the starting position is guided relative to the first bearing point, wherein the trigger element is adapted to exert a force on the locking element at an angle to the guideway.
11. The crash device according to claim 1, wherein the carrier forms a housing in which the arresting element is arranged.
12. The crash device according to claim 1, wherein the carrier includes two housing sheets welded to each other.
13. The crash device according to claim 12, wherein the guide is configured in the form of an oblong hole in at least one of the housing sheets.
14. The crash device according to claim 1, wherein the second bearing point comprises a bearing bush which in the starting position is pressed into the arresting element.
15. The crash device according to claim 1, wherein at least one of the first bearing point and the second bearing point each are configured for rotatable mounting.
16. The crash device according to claim 1, wherein unlocking of the locking device is irreversible.
17. A vehicle seat comprising a seat part and a backrest, and comprising at least one crash device according to claim 1.
18. The vehicle seat according to claim 17, wherein the seat part is supported on a base via the crash device.
19. The vehicle seat according to claim 18, wherein the crash device is pivotally mounted on the base with one of the bearing points, and the seat part is pivotally mounted on the other one of the bearing points.
20. The vehicle seat according to claim 19, wherein the seat part is height-adjustably mounted relative to the base via the crash device.